Mohib Amin

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WORK EXPERIENCE

SQL Developer Co-op | Environment Climate Change Canada Jun. 2024 - May. 2025

- Designed and optimized complex **PL/SQL** queries and database objects within an **Oracle** relational system, reducing average query execution time by **25%** through indexing and execution plan tuning.
- Managed relational database objects including tables, views, indexes, and schemas supporting production datasets exceeding **100K+ records**, improving system responsiveness and ensuring data integrity across environments.
- Collaborated with developers to troubleshoot database performance bottlenecks and implement access control, role management, and data security best practices.

Software QA Co-op | Environment Climate Change Canada May. 2025 - Sept. 2025

- Implemented automated testing workflows using **Robot Framework**, executing **50+ regression test cases per release cycle** and increasing **automated** coverage by **30%** across evolving feature releases.
- Used **Azure DevOps** to manage bugs, user stories, and comprehensive test plans across multiple **sprints**, improving **defect** traceability and contributing to more stable and reliable release cycles.

EDUCATION

Bachelor of Engineering in Computer Engineering
Toronto Metropolitan University (Formerly known as Ryerson University) Sept. 2021 - Apr. 2026

SKILLS

Programming Languages: Java, HTML, C#, CSS, MATLAB, C, JavaScript, SQL, JavaFX, Python, VHDL, C++
Platform/Tools: Git, IntelliJ, Linux, Devops, Jira, Arduino, UNIX, LLM, ML Agents, Pytorch, Unity, Ubuntu

RELEVANT PROJECTS

Smart Parking Pathfinder (Senior Design Project - 1st Place) | *Arduino, Python (Flask, Socket.IO), JavaScript, HTML, CSS*

- Architected a real-time smart parking solution connecting **Arduino**-based sensors, LED indicators, and servo-controlled gates to a live, browser-based parking map for continuous occupancy monitoring.
- Integrated a **Python** backend using **Flask** and **Socket.IO** to process serial data from hardware and broadcast real-time parking occupancy updates with **sub-second (200ms) latency** to connected clients.
- Constructed an interactive multi-level parking UI showcasing live availability, EV/accessibility classifications, and dynamic counts using responsive front-end components.
- Validated system behavior by implementing a simulation mode and manual update **API** to test real-time performance, verify data flow integrity, and support efficient debugging without physical hardware.

AI-Powered NPC Behavior Modeling in Unity | *Unity, C#, ML-Agents, Reinforcement Learning, PPO, Pytorch*

- Built reinforcement learning NPCs in **Unity** using **ML-Agents** with custom observation/action spaces and reward design to enable adaptive navigation, combat decision-making, and context-aware behavioral responses within dynamic real-time gameplay environments.
- Improved agent convergence stability by approximately **30%** through iterative training cycles and structured gameplay evaluation, systematically tuning reward parameters and behavior policies to reduce erratic actions and outperform traditional scripted NPC logic in complex in-game scenarios.

Discord Job Automation Bot | *Python, Playwright, Discord API, Ubuntu VM*

- Engineered and deployed a **Python**-based Discord bot to automate job postings across multiple dynamic career platforms, processing **100+ listings per day** using **Playwright** for headless browser automation and delivering structured real-time updates via **Discord webhooks**.
- Deployed the bot on a remote **Ubuntu VM** (SSH-based) for persistent **24/7** background execution, implementing replay protection, deduplication logic, dependency management, and automated restart processes to maintain high availability and reliable job alerts.